

Nonnegative and positive factorizations

# Exact positive semidefinite rank of subset-intersection matrices

**Difficulty:** challenging **Importance:** interesting to specialist

**Status:** Partially resolved

**Rating rationale:** Challenging because the symmetric subset structure has not yielded matching PSD-rank bounds; specialist importance reflects this explicit family's role as a benchmark for semidefinite factorization.

**Area:** structured positive semidefinite factorization

## Further partial results — 2026-09-13

**Author:** Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. **Independent Codex AI-agent review:** PASS for the new partial scope.

For hypothetical real size-four factorizations, at most 65 of 70 factors on either side have rank one at order eight, and at most 64 of 126 at order nine. A span-five theorem for projection-valued subset sums gives mixed-rank and 252 covering constraints on the affine factor side at order nine. Exact integer and finite-field calculations establish a 90-dimensional quartic first-derivative space; a hypothetical determinant must use its nine extra directions. See Sections 2–10 of the [authored manuscript \(TeX\)](#).

**Remaining question:** The exact real PSD ranks for all  $n \geq 7$  remain undetermined, including orders seven, eight and nine. These necessary conditions do not exclude every size-four factorization. The stronger prior draft bounds quoted in the manuscript are outside this independent review's scope and are not newly endorsed here. Status remains **Partially resolved**.

[Independent review and exact checks](#) · [Submission, verified affiliation and provenance](#). The review is informal AI-agent review; no Lean verification or external human peer review is asserted.

## Partial result — 2026-09-11

**Author:** Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review:** PASS for the stated partial scope.

The real positive semidefinite rank is exactly four for  $n = 5$  and  $n = 6$ . Explicit graph factors give the general bound  $\text{rank}_{\text{psd}} M^{(n)} \leq \lceil 2\sqrt{2}[(n-1)/2] \rceil$ , and submatrix monotonicity gives a lower bound of four for every  $n \geq 5$ .

**Remaining question:** The exact ranks as a function of  $n$  remain undetermined for  $n \geq 7$ . In particular, the new upper bound five at  $n = 7, 8$  is not accompanied by a matching lower bound five. The finite orders remain part of this single family entry. The ratings assess that surviving question.

**Primary reference:** [complete authored PDF](#), [standalone TeX](#), [Theorem 1](#), [Corollary 5](#) and [equation \(8\)](#). [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

## Context and notation

Write  $\mathbb{S}_+^k$  for the cone of real symmetric positive semidefinite  $k \times k$  matrices. For an entrywise nonnegative matrix  $M \in \mathbb{R}_+^{p \times q}$ , its **real positive semidefinite rank** is

$$\text{rank}_{\text{psd}}(M) = \min\{k \geq 1 : \exists A_1, \dots, A_p, B_1, \dots, B_q \in \mathbb{S}_+^k, \quad M_{ij} = \text{tr}(A_i B_j) \text{ for all } i, j\}.$$

This definition concerns a family of matrix factors indexed by the rows and columns of  $M$ .

### Problem statement

For each integer  $n \geq 5$ , let  $\mathcal{I}_n$  and  $\mathcal{J}_n$  consist of the subsets of  $\{1, \dots, n\}$  of cardinalities  $\lfloor n/2 \rfloor$  and  $\lceil n/2 \rceil$ , respectively. Define the nonnegative matrix

$$M^{(n)} \in \mathbb{R}_+^{\mathcal{I}_n \times \mathcal{J}_n}, \quad M_{I,J}^{(n)} = |I \cap J|.$$

### Question

Determine  $\text{rank}_{\text{psd}}(M^{(n)})$  exactly as a function of  $n$ , with real symmetric factors as defined above. The family is one problem; its individual orders are not separate catalog entries.

The source supplies the bounds

$$\left\lceil \frac{\sqrt{1+8n}-1}{2} \right\rceil \leq \text{rank}_{\text{psd}}(M^{(n)}) \leq 2\lceil \sqrt{n} \rceil.$$

These matrices provide structured benchmarks for algorithms seeking small positive semidefinite factorizations.

### References

Hamza Fawzi, João Gouveia, Pablo A. Parrilo, Richard Z. Robinson, and Rekha R. Thomas, *Positive semidefinite rank*, Mathematical Programming **153** (2015), 133–177, §9.1, Problem 9.2. Arnaud Vandaele, François Glineur, and Nicolas Gillis, *Algorithms for Positive Semidefinite Factorization*, Computational Optimization and Applications **71** (2018), 193–219, §4.2, preprint pp. 10–11.

### Status check — 2026-09-10

Rechecked [Fawzi et al., Problem 9.2](#) and [Vandaele–Glineur–Gillis, §4.2](#), and searched by subset intersections, Johnson schemes and the problem number. The sources support the displayed real-factor bounds but not an exact formula. No later determination was located; the 2018 algorithm paper remains the latest explicit status source checked, so the later evidence is limited to this search.